Presentation Script & Defence Guide

Hack4Dev - Building age classification from satellite imagery - Chiwuswa Joseph

Timing plan (5 minutes, 8 slides)

Slide	Topic	Time	Finish by
1	Title	15 s	0:15
2	Abstract	30 s	0:45
3	Data preparation and Madrid model	45 s	1:30
4	Confusion matrix	30 s	2:00
5	Transfer from Madrid to Amsterdam	45 s	2:45
6	Transfer methods compared (chart)	30 s	3:15
7	**Macro F1 results and learning curve (key slide)**	60 s	4:15
8	Design decisions and conclusion	45 s	5:00

Numbers to remember

What	Value
Madrid macro-F1 (5-fold spatial CV)	**0.656 +/- 0.030**
Amsterdam, 5 labels per class	0.605 +/- 0.063
Amsterdam, 25 labels per class (low-data prize)	**0.685 +/- 0.021**
Amsterdam, 50 / 100 labels per class	0.712 / 0.733
Amsterdam, 200 labels per class (peak)	**0.739 +/- 0.005**
Average Amsterdam F1 (prize 1)	**0.695** (baseline 0.599)
Zero-shot Amsterdam (no labels)	0.559 (baseline 0.343)
Random Forest baseline, Madrid, same folds	0.606
Number of features	277
Pixels	Madrid 76,263 - Amsterdam 25,992

If you run short of time: keep slide 7 in full, shorten slides 4 and 6 to one sentence each.

What to say - slide by slide

Slide 1 - Title (15 s)

"Good morning. I am Chiwuswa Joseph. My project classifies the age of buildings from satellite images, and transfers a model trained in Madrid to Amsterdam using very few labels."

Slide 2 - Abstract (30 s)

"In short: I use 40 years of Landsat data for every 30-metre pixel, turn it into features that work across cities, and add information about each pixel's neighbourhood. With only 25 labelled pixels per class, Amsterdam reaches a macro-F1 of 0.685, and the model beats the baseline at every sample size."

Slide 3 - Data preparation and Madrid model (45 s)

"First, I removed cloudy, saturated and hazy observations and combined each summer's scenes with the median. The labels are the four age classes, built from the construction year. Each pixel's 40-year history becomes 277 features - for example decade averages, the year the surface suddenly changed, which is usually the construction year, and averages over the surrounding neighbourhood. Gradient boosting beat Random Forest on the same folds. I used spatial cross-validation, so nearby pixels cannot appear in both training and testing. Madrid macro-F1 is 0.656, against 0.606 for the baseline."

Slide 4 - Confusion matrix (30 s)

"Most predictions are on the diagonal, so they are correct. The main mistake is class 1 versus class 2: both were built before satellites started recording in 1984, so there is no construction moment to detect. Mistakes are almost always between neighbouring classes - the model rarely confuses very old with very new."

Slide 5 - Transfer method (45 s)

"To move to Amsterdam, I combine two models. Model A re-uses what Madrid learned and only recalibrates it. Model B learns from the few Amsterdam labels. With few labels, A gets 40 percent of the weight; from 100 labels per class, B does it alone. For a fair comparison, every method was tested on exactly the same random samples as the baseline, with a fixed random seed."

Slide 6 - Transfer methods compared (30 s)

"The red line is my combined model. It is the highest at every sample size - combining Madrid's knowledge with local data beats either one on its own."

Slide 7 - Results and learning curve (60 s) - KEY SLIDE, slow down

"Here are the main results. Madrid scores 0.656. In Amsterdam, macro-F1 rises from 0.605 with just 5 labels per class to 0.739 with 200. The curve rises fastest between 5 and 50 labels and then flattens, so about 50 to 100 labels per class is enough. The grey line is the baseline: my model is better at every point - 0.695 on average against 0.599. And with only 25 labels, Amsterdam already scores higher than Madrid."

Slide 8 - Conclusion (45 s)

"What worked best was spatial context, because buildings of the same age are built together. Smoothing the predictions with their neighbours also helped every time. What did not work: predicting the year directly, the year-to-year change features, and a very large 900-metre neighbourhood - so I removed them. The hardest case remains class 1 versus class 2. Next, I would use the unlabelled Amsterdam pixels for self-training. Thank you - I am happy to take questions."

Likely questions from the judges - and short answers

Answer in one or two sentences, then stop. If you do not know, say so honestly and say what you would check.

The problem

Q: What problem are you solving?

Predicting when buildings were built (4 age classes) from satellite data, and making the model work in a new city with only a few labelled examples.

Q: Why does it matter?

Building age helps with energy planning, safety and urban planning, but many cities have no records. Satellite data is free and covers the whole world.

Q: Why macro-F1 and not accuracy?

Macro-F1 gives every class equal weight. With accuracy, a model could score well just by predicting the biggest class.

Q: What does the +/- mean?

The spread of the score over the different test splits or random samples. Small means the result is stable.

Data and features

Q: What is one data sample?

One 30 x 30 m pixel. Its 42 years of satellite values are summarised into one row of features.

Q: How did you clean the data?

I removed cloudy, saturated and hazy observations (Blue above 15,000), combined each year's scenes with the median, and filled missing years by interpolation.

Q: How did you create the labels?

From the construction year with pd.cut: before 1960 (Madrid) or 1945 (Amsterdam), up to 1984, 1984-2004, and after 2004.

Q: Why normalise per city and year?

Madrid and Amsterdam differ in climate, materials and sensors. Comparing each pixel with its own city's median removes those differences, so the features mean the same thing in both cities.

Q: Does that use Amsterdam data?

Only the unlabelled satellite values, never the age labels - this is standard domain adaptation.

Q: What is a change-point feature?

The year where a pixel's values suddenly change - usually when a building was constructed. It is what separates class 3 from class 4.

Q: Why neighbourhood features?

Buildings of the same era are built together, in old centres or new estates. Averages over 90 m and 300 m add that context. They gave the biggest improvement.

Q: 277 features - isn't that too many?

Gradient boosting picks the useful ones itself. I tested keeping fewer features and the score stayed the same, and I removed the 33 features that had negative importance.

Model and validation

Q: Why gradient boosting?

On the same folds it scored highest: 0.622 against Random Forest 0.606, ordinal 0.618 and regression 0.568.

Q: How did you handle class imbalance?

I divided the predicted probabilities by how common each class is, so the large class 2 is not over-predicted.

Q: Why spatial cross-validation?

Neighbouring pixels are often the same building. A random split would let the model see its neighbours and give an over-optimistic score. Whole 1.5 km blocks go to the same fold.

Q: The organisers' baseline scored 0.628 - is yours really better?

That 0.628 used random cross-validation, which is easier. On the same spatial folds the baseline scores 0.606, and mine 0.656.

Q: Is there any data leakage?

No. The construction year is never a feature, there is one row per pixel, neighbourhood features use only satellite values, and smoothing only uses pixels from the same test fold.

Transfer to Amsterdam

Q: Explain the transfer simply.

Model A reuses Madrid's knowledge and just recalibrates it. Model B learns from the Amsterdam labels. I mix them: few labels means trust Madrid, many labels means trust Amsterdam.

Q: How did you choose the weights without cheating?

I tuned them on separate random samples, then reported scores on different samples - the same ones used for the baseline. The organisers also test on held-out data I never saw.

Q: What is the spatial smoothing?

Each pixel's prediction is mixed 50/50 with the average of its 5 x 5 pixel neighbourhood. It uses no extra labels.

Q: Why is the error bar so large at 5 labels?

With only 20 labelled pixels, the result depends a lot on which pixels are drawn. It shrinks quickly as n grows.

Q: Why does the curve flatten after 100 labels?

By then the model has learned Amsterdam's patterns; extra labels add little new information.

Q: Why is zero-shot only 0.559?

Without any labels the model cannot know Amsterdam's own boundary (1945 vs 1960) or its building styles. It is still much better than the baseline's 0.343.

Limitations and next steps

Q: What is the biggest weakness?

Class 1 versus class 2 - both are older than the satellite record, so there is no construction signal. Also, labels are averages over mixed 30 m pixels, so they are noisy.

Q: What did not work?

Predicting the year directly (zero-shot 0.39), the prototype method (weight 0), the year-to-year change features, and a 900 m neighbourhood (more features, no gain).

Q: Would it work in another city, like Nairobi?

The idea transfers - city-relative features and a few local labels - but it is only tested on Madrid to Amsterdam. Very different building materials would need checking.

Q: Is it reproducible?

Yes. Fixed seed 42, the notebooks run end to end, and the Madrid model and all five Amsterdam models are saved.

Q: What would you do with more time?

Self-training on the unlabelled Amsterdam pixels, a deep model on the full time series, and testing more cities.

Q: Did you use AI tools?

Yes, as a coding assistant. I ran every experiment myself, checked the results, and can explain each step.

Delivery tips

- Look at the judges, not the slides; point at the chart when you mention a number.
- Say each key number once, clearly: 0.656, 0.685, 0.739, 0.695.
- If asked something you do not know: "Good question - I did not test that; I would check it by..."
- Finish on time: end with the conclusion slide and "Thank you".